

Qatar-2b

R-1627

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_190307 · created 2026-08-02T05:18:16+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458549.87929 +/- 0.00078 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.228 +/- 0.066
Transit depth (Rp/Rs)^2	5.18 +/- 3.0 [%]
Orbital Inclination (inc)	89.71 +/- 1.74
Airmass coefficient 1 (a1)	5.88 +/- 2.65
Airmass coefficient 2 (a2)	-1.07 +/- 0.63
Scatter in the residuals of the lightcurve fit is	58.58 %
Best Comparison Star	#1 - [608, 420]
Optimal Aperture	1.23
Optimal Annulus	11.5
Transit Duration (day)	0.0805 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.228 ± 0.066 vs 0.16327 (deviation 0.88σ)
Duration fit vs published	0.0805 d vs 0.0767 d (deviation 0.73σ)
Mid-time O–C	1.7 min
Depth SNR	3.1
Residual scatter	58.58 %

Light curve

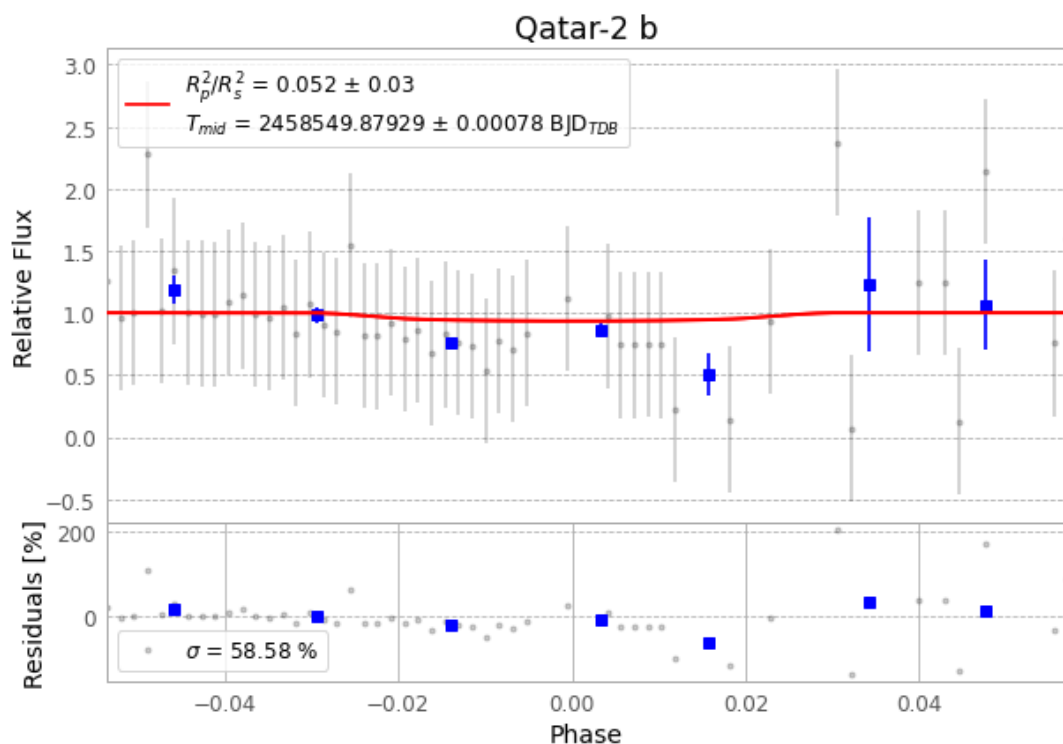


Figure 1 — FinalLightCurve_Qatar-2 b_2019-03-07.png (SHA-256 18f694f36f9dac03...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [508, 61], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3f2bd525a703d1f1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), Qatar-2190307071513.FITS → Qatar-2190307110312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-190307.log (58e7028d76b2...), FinalLightCurve_Qatar-2 b_2019-03-07.png (18f694f36f9d...), FinalLightCurve_Qatar-2 b_2019-03-07.csv (24d8dd91d77d...)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 05:18Z.